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CS23532-COMPUTER NETWORKS-LAB MANUAL

PRACTICAL – 11 (a) & 11 (b)

AIM:

To simulate and configure Static Routing and RIP (Routing Information Protocol) in Cisco Packet Tracer, establish communication between different networks, and observe how routers forward packets at the network layer.

PART (a): Simulate Static Routing Protocol Configuration Using Cisco Packet Tracer

1. **Design the Network Topology:**
   Create an internetwork using **3 routers, 2 switches, and 4 PCs**. Connect the routers in a line so that static routes can be configured manually.

   Using the following devices:

- 3 Routers — Cisco 2911
- 2 Switches — Cisco 2960
- 4 PCs
- Ethernet cables and serial cables, if supported by the router models

2. **Assign IP Addresses:**
   Configure the PCs through PC → Desktop → IP Configuration.
   Default gateways:

- PC0 and PC1: 192.168.1.1
- PC2 and PC3: 192.168.2.1

3. Configure the Router Interfaces:

The following example uses Ethernet interfaces for the router-to-router links. If your topology uses serial interfaces, substitute the correct interface names and use no shutdown. For serial DCE interfaces, configure a clock rate if required.

Terminal Command:

- **Router R0:**

```
enable
configure terminal

interface gigabitEthernet 0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit

interface gigabitEthernet 0/1
ip address 10.0.12.1 255.255.255.252
no shutdown
exit

end
```

- **Router R1:**

```
enable
configure terminal

interface gigabitEthernet 0/0
ip address 10.0.12.2 255.255.255.252
no shutdown
exit

interface gigabitEthernet 0/1
```

ip address 10.0.23.1 255.255.255.252
no shutdown
exit

end

- **Router R2:**

enable
configure terminal

interface gigabitEthernet 0/0
ip address 10.0.23.2 255.255.255.252
no shutdown
exit

interface gigabitEthernet 0/1
ip address 192.168.2.1 255.255.255.0
no shutdown
exit

end

Verify the interfaces on each router: show ip interface brief

4. **Configure Static Routes:**

Static routes are manually entered into each router's routing table. They tell the router how to reach networks that are not directly connected.

Terminal Command:

- **Router R0:**

enable

configure terminal
ip route 192.168.2.0 255.255.255.0 10.0.12.2
end

- **Router R1:**

enable
configure terminal
ip route 192.168.1.0 255.255.255.0 10.0.12.1
ip route 192.168.2.0 255.255.255.0 10.0.23.2
end

- **Router R2:**

enable
configure terminal
ip route 192.168.1.0 255.255.255.0 10.0.23.1
end

- **Verify the routing table on each router:** show ip route

5. Test Communication Between Networks:

- **Open PC0 → Desktop → Command Prompt and type:** ping 192.168.2.10

Outcome:

Packets: Sent = 4
Received = 4
Lost = 0

6. Simulate Packet Transmission:

- Select **Simulation** mode.
- Select **Add Simple PDU**.
- Click PC0 and then PC2.
- Click **Auto Capture / Play**.
- Observe the packet passing through R0, R1, and R2 before reaching PC2.

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PART (b): Simulate RIP Using Cisco Packet Tracer

1. **Design the Network:**

Use the same topology and IP addressing scheme from Part (a). This allows you to compare static routing with dynamic routing.

RIP automatically shares network routes between routers, reducing the need to configure each remote route manually.

2. **Configure RIP on Router R0:**

Terminal Commands:

```
enable
configure terminal
router rip
version 2
no auto-summary
network 192.168.1.0
network 10.0.0.0
end
```

3. **Configure RIP on Router R1:**

Terminal Commands:

```
enable
configure terminal
router rip
version 2
no auto-summary
network 10.0.0.0
end
```

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4. **Configure RIP on Router R2:**

   Terminal Commands:

   enable

   configure terminal

   router rip

   version 2

   no auto-summary

   network 192.168.2.0

   network 10.0.0.0

   end

5. **Verify RIP Routing:**

   Terminal Commands:

   show ip route

   show ip protocols

6. **Test RIP Communication:**

   Terminal Commands:

   ping 192.168.2.10

   Expected Outcome:

   Packets: Sent = 4

   Received = 4

   Lost = 0

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7. Simulate RIP Packet Routing:

- Switch to Simulation mode.
- Use Add Simple PDU to send a packet from PC0 to PC2.
- Click Auto Capture / Play.
- Observe the packet travelling through R0, R1, and R2.

RESULT:

Thus, the experiment demonstrated static routing, dynamic routing using RIP, routing table verification, and packet forwarding at the network layer.